G. Workflow & State-Machine Infrastructure

Standalone capability brief · Suggested module: **Workflow Runtime**

Purpose

Run long-lived, multi-step business and infrastructure processes reliably across services, including retries, human approvals, compensation, and recovery after failures.

1. Scope and capability areas

Durable orchestration

Persist workflow state and resume execution after process or infrastructure restarts.

Human approvals

Model approvals, rejections, timeouts, delegation, and audit history.

Versioning and migration

Keep active workflows compatible as definitions change; provide explicit migration paths.

Compensation and sagas

Define compensating actions for partial completion where true distributed transactions are unavailable.

Concurrency and timeouts

Coordinate concurrent operations, enforce deadlines, and prevent duplicate execution.

Operational visibility

Expose workflow status, step history, retry state, owner, and recovery actions.

2. Shared contracts and standards

- Each workflow declares inputs, outputs, version, timeout, retry policy, idempotency strategy, and compensation behavior.
- Activities are designed to be idempotent because retries may repeat execution.
- Human approvals and administrative overrides are auditable.
- Workflow definitions distinguish retryable, terminal, and compensatable failures.

3. Recommended implementation approach

- Use a durable workflow engine or equivalent persisted state-machine runtime rather than in-memory orchestration for critical processes.
- Define activity contracts and isolate external side effects behind idempotent adapters.
- Provide operational tooling for pause, resume, retry, cancel, and inspect, with role-based permissions.
- Test process crashes, duplicate signals, late callbacks, and definition upgrades.
- Use workflow-specific dashboards and alerts for stuck or repeatedly failing executions.

4. Failure behavior and operational expectations

- Retries use bounded exponential backoff with jitter and a maximum attempt or time budget.
- Compensation failures remain visible and require retry or operator intervention; do not mark the workflow successful.
- Unknown workflow versions must be quarantined or handled through a documented migration path.

5. Suggested operational metrics

- Workflow completion and failure rates
- Age of oldest running workflow
- Retries per activity
- Compensation failure rate
- Time spent awaiting human approval

6. Readiness and acceptance checklist

- ■ Critical workflows resume after worker or service restarts.
- ■ Duplicate activity execution does not create duplicate business effects.
- ■ Operators can inspect and safely recover stuck workflows.
- ■ Workflow version changes have tested migration behavior.

Ownership note: assign a named team, service owner, support route, versioning policy, and deprecation process before treating this capability as a platform contract.